

The Pair-Creation Threshold of Impulsive Stueckelberg–Horwitz–Piron Scattering Is a Cayley-Transform Pole

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In classical Stueckelberg electrodynamics a particle worldline represents pair annihilation when its time coordinate reverses direction with respect to the evolution parameter. A recent calculation by Land exhibits this reversal in impulsive Coulomb scattering and finds a sharp threshold, the reversal occurring exactly when a dimensionless interaction strength exceeds two, equivalently when the interaction energy exceeds the rest energy of the annihilated pair. We replicate the published calculation exactly and then locate the origin of its threshold. The impulsive midpoint convention used to close the scattering equations is, for the reduced two-dimensional kick system, precisely the Cayley transform of the interaction generator, which is the lowest diagonal Padé approximant of the exponential map. The published threshold is the pole of that approximant. Integrating the same generator continuously instead gives hyperbolic evolution that conserves a quadratic form and provably never reverses the time velocity from physical initial data, at any interaction strength. In group language, the continuous flow stays in the identity component of the invariance group of the conserved form, while beyond the pole the Cayley transform lands in the disconnected component containing the simultaneous reversal of time velocity and radial velocity, which no continuous flow of the generator can reach. The published algebra is correct as published. What our result shows is that its classical pair event marks the breakdown radius of an approximant rather than a dynamical transition of the continuously accumulated force, and that whether the full field dynamics restores a genuine threshold is an open question. A practical corollary for simulation work is that reversal statistics can be created by a discretization prescription alone.

I. INTRODUCTION

Stueckelberg proposed that a single classical worldline can describe pair annihilation if its laboratory time coordinate, viewed as a function of an independent evolution parameter, turns around [1, 2]. The framework of Horwitz and Piron gives this idea a covariant Hamiltonian mechanics [3], and its gauge-invariant electrodynamics permits the mass exchange with the field that a turnaround requires [5]. Within that theory, Land computed a concrete example [4]. An event scatters impulsively in the Coulomb field of a heavy nucleus, and its outgoing time velocity reverses sign exactly when a dimensionless coupling, built from the correlation length, the interaction distance, and the charges, exceeds two. The condition translates to the statement that the interaction energy exceeds the rest energy of the annihilated pair, which is physically appealing, and the calculation is fully explicit.

This paper reports a replication of that calculation and one structural finding about it. The replication is exact. Every printed equation we transcribe passes machine-precision consistency guards, the threshold appears exactly where the source places it, and the limiting outgoing energy agrees with the printed asymptote. The finding concerns where the threshold comes from. The source closes its impulsive scattering equations with a midpoint convention, evaluating the interaction with the average of the incoming and outgoing velocities. For the reduced system that governs the time velocity and the

radial velocity component, that convention is exactly the Cayley transform of the interaction generator. The Cayley transform is the simplest diagonal Padé approximant of the exponential map, and it has a pole. The published threshold sits exactly on that pole. The alternative closure, continuous accumulation of the same generator, has no pole. It conserves a quadratic form that confines the motion to one branch of a hyperbola, and from physical initial data it never reverses the time velocity at any coupling strength.

Three claim types appear below and are kept separate. Statements about the source’s published algebra are replication results, verified to machine precision. The identification of the midpoint closure with the Cayley transform, the conservation law of the continuous closure, and the component structure of the two maps are proved. Statements about which closure the full field theory prefers are open questions, and we make no claim about physical pair production anywhere in this paper.

II. THE REPLICATED CONFIGURATION

We follow the notation of Ref. [4] and use its equation numbers. An event with coordinates $x^\mu(\tau)$ evolves under the Lorentz force of five τ -dependent gauge fields (source Eq. 12), and may exchange mass with the field (source Eq. 14), which is what permits $\dot{t} = dt/d\tau$ to change sign. The incoming trajectory has velocity $u = \dot{t}_{\text{in}}(1, v, 0, 0)$ with $\dot{t}_{\text{in}} = 1/\sqrt{1-v^2}$ (source Eqs. 47–48). The nucleus induces the potential $a^0 = a^5 = (Ze/4\pi R)\delta(\tau - \tau_1)$ at interaction distance R in the small-correlation-length limit

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(source Eq. 56). The strength of the kick is the dimensionless coupling

$$g_e = \frac{\lambda}{M} \frac{Ze^2}{4\pi R^2}, \quad (1)$$

with λ the correlation length and M the Galilean target mass (source Eq. 64). Closing the impulse integrals with the midpoint velocities (source Eq. 63) yields a linear system (source Eq. 66) whose solution (source Eq. 67) has time component

$$\dot{t}_f = \frac{\dot{t}_{\text{in}} \left(1 - g_e v \hat{R}_x\right) + \frac{1}{4} g_e^2 (\dot{t}_{\text{in}} + 2)}{1 - \frac{1}{4} g_e^2}, \quad (2)$$

where $\hat{R}_x$ is the component of the unit vector to the interaction point along the incoming motion (source Eq. 76). The numerator is positive definite (source Eq. 78), so the sign of $\dot{t}_f$ is the sign of the denominator, giving reversal exactly for $g_e > 2$ (source Eqs. 79–80), and $\dot{t}_f \rightarrow -(\dot{t}_{\text{in}} + 2)$ as $g_e \rightarrow \infty$ (source Eq. 81).

Our transcription of the system and of the closed form agree with each other to 5.7×10^{-14} over a ninety-cell parameter sweep, the time component of the closed form matches Eq. (2) to 1.4×10^{-14} , the threshold is exact in every cell, and the asymptote is reproduced to 1.4×10^{-3} at $g_e = 1000$. The replication record, with a line-by-line provenance note mapping each source equation to code, is part of the repository described in Sec. VI.

III. THE REDUCED KICK SYSTEM AND ITS TWO CLOSURES

During the kick the source evaluates the potential at the fixed interaction point, which is what its delta collapse does in source Eq. 61. Writing w for the component of the spatial velocity along $\hat{R}$ and $s \in [0, 1]$ for the accumulated interaction kernel, the force equations (source Eqs. 59–60) reduce to the linear system

$$\frac{d}{ds} \begin{pmatrix} \dot{t} + 1 \\ w \end{pmatrix} = G \begin{pmatrix} \dot{t} + 1 \\ w \end{pmatrix}, \quad G = -g_e \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (3)$$

with the perpendicular velocity component unchanged. The kernel-derivative part of the force is a total derivative whose contribution vanishes identically in the variable s . Everything below is a statement about the two ways of turning the generator G into a map from incoming to outgoing velocities.

Proposition 1 (Identification). *The midpoint closure of source Eq. 63 applied to the reduced system (3) is the Cayley transform $v_f = (I - G/2)^{-1}(I + G/2)v_{\text{in}}$, and its time component reproduces Eq. (2) identically.*

Proof. The midpoint convention evaluates the velocity in the accumulated kick as the average of the endpoint velocities, so $v_f - v_{\text{in}} = G(v_{\text{in}} + v_f)/2$, which rearranges

to the stated transform. Expanding with $G^2 = g_e^2 I$ gives $v_f = [(1 + \frac{1}{4}g_e^2)I + G]v_{\text{in}}/(1 - \frac{1}{4}g_e^2)$, whose first component is Eq. (2) after substituting $w_{\text{in}} = v_{\text{in}}\hat{R}_x$. The identity was also verified numerically to 10^{-12} across the sweep of Sec. IV. $\square$

The Cayley transform is the Padé approximant of order $(1, 1)$ of the exponential map [6, 7]. Its eigenvalue action on the eigenvectors of G , which have eigenvalues $\mp g_e$, is $\lambda \mapsto (1 + \lambda/2)/(1 - \lambda/2)$, so one image eigenvalue is $(1 + g_e/2)/(1 - g_e/2)$. The pole at $g_e = 2$ is the published threshold.

Proposition 2 (Conservation and no reversal). *The continuous flow $\exp(sG)$ conserves $Q = (\dot{t} + 1)^2 - w^2$. For physical initial data, meaning $\dot{t}_{\text{in}} + 1 > |w_{\text{in}}|$, the continuously accumulated kick satisfies $\dot{t}(s) + 1 \geq \sqrt{Q} > 0$ for all s and all g_e , so the time velocity never reverses.*

Proof. G satisfies $G^T \eta + \eta G = 0$ for $\eta = \text{diag}(1, -1)$, so $\exp(sG)$ lies in the invariance group of Q . In light-cone coordinates $(\dot{t} + 1) \pm w$ the flow is diagonal with factors $e^{\mp g_e s}$, so their product Q is constant and both coordinates remain positive if they start positive. Incoming data has $\dot{t}_{\text{in}} + 1 > 1 > |w_{\text{in}}|$ whenever $|v| < 1$, since $|w_{\text{in}}| \leq v\dot{t}_{\text{in}} < \dot{t}_{\text{in}}$. The closed form $\dot{t}(s) + 1 = (\dot{t}_{\text{in}} + 1) \cosh(g_e s) - w_{\text{in}} \sinh(g_e s)$ was verified against an independent fixed-step integration to 10^{-9} . $\square$

Proposition 3 (Component structure). *The invariance group of Q has disconnected components. For $g_e < 2$ the Cayley transform lies in the identity component and maps the physical branch $\dot{t} + 1 \geq \sqrt{Q}$ to itself. For $g_e > 2$ both of its eigenvalues are negative, so it is the product of a boost with the total reflection $-I$, lands in the component that reverses both $\dot{t} + 1$ and w , and maps the physical branch to the opposite branch $\dot{t} + 1 \leq -\sqrt{Q}$. No continuous one-parameter flow of G reaches that component.*

Proof. The image eigenvalues $(1 \mp g_e/2)/(1 \pm g_e/2)$ are both positive for $g_e < 2$ and both negative for $g_e > 2$, with product one in both cases, so the map is a Q -preserving matrix equal to a boost for $g_e < 2$ and to $-I$ times a boost for $g_e > 2$. The exponential of a fixed generator is connected to the identity, so it cannot equal a map with negative eigenvalues. Branch preservation and branch exchange follow from the sign of the eigenvalues on the light-cone coordinates. $\square$

IV. NUMERICAL VERIFICATION

All numbers are from the committed evidence record of the replication run. The transcription guards give agreement between the printed linear system and the printed closed form at 5.7×10^{-14} and between the closed form and its time component at 1.4×10^{-14} . The continuous closed form of Proposition 2 agrees with an independent Runge–Kutta integration to better than 10^{-9} at couplings one,

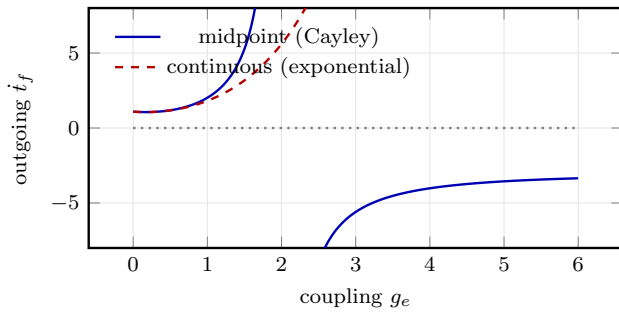


FIG. 1. Outgoing time velocity against coupling for the reference scattering geometry. The midpoint closure diverges at the pole $g_e = 2$ and emerges reversed, which is the published pair event. The continuous closure of the same generator grows monotonically and never reverses. Both curves are computed from the committed replication code.

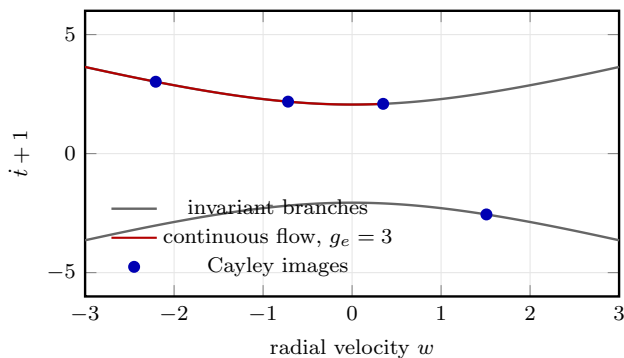


FIG. 2. The conserved-form picture. Physical initial data sits on the upper branch of the invariant hyperbola. The continuous flow moves along that branch and cannot leave it. The Cayley images move along the upper branch as the coupling grows toward two, then reappear on the lower branch beyond the pole. Every plotted Cayley image satisfies the invariant to 10^{-9} , verified in the committed data script.

three, and six. At the reference geometry with $v = 0.4$ and $\hat{R}_x = 0.8$ the two closures give outgoing time velocities of 2.02 against 1.82 at coupling one, -5.60 against $+16.55$ at coupling three, and -3.35 against $+350.4$ at coupling six, the midpoint values approaching the printed asymptote from below while the continuous values grow exponentially. The sweep also finds spacelike outgoing velocities in thirty percent of its cells at intermediate coupling, which is not an error, since a worldline that reverses its time direction must cross the spacelike region, and the source's timelike statement applies to its large-coupling limit.

V. WHAT IS AND IS NOT IMPLIED

The published calculation is internally exact and our replication confirms every printed consequence of its stated conventions. The midpoint convention is, moreover, not a numerical shortcut that the source could simply have avoided. It is the closure that defines the impulsive model, adopted for the same reason implicit midpoint methods are prized in geometric integration, and the resulting map does preserve the invariant form of the reduced system exactly [6]. The point established here is narrower and, we think, sharper. Within the impulsive model the pair threshold is located exactly at the pole where the Cayley approximant of the kick exponential leaves the identity component of the invariance group, and the continuously accumulated version of the same kick provably never produces a reversal. The classical pair event of this configuration is therefore a property of the closure prescription, not of the accumulated force.

Whether that observation deflates the physical picture or refines it is an open dynamical question that the impulse limit cannot settle. The full theory evolves the potential with the moving event, includes radiation, and determines the interaction window self-consistently, and none of that structure survives in the frozen-point kick. A configuration in which the full pre-Maxwell field dynamics produces or forbids a reversal threshold would decide the question, and we state it as such rather than prejudge it.

There is also a practical corollary for any simulation campaign that counts time reversals, including ours. A discretization prescription can manufacture a reversal threshold at the breakdown radius of its own approximant, complete with a physically appealing energy condition, while the continuous dynamics it approximates has none. Reversal statistics inherit the component structure of the map used to close each interaction, so a campaign that measures them must either integrate continuously through interactions or preregister its closure prescription and test sensitivity to it. This lesson is now part of the anti-circularity contract of the pair-production campaign that produced this replication.

VI. METHODS AND REPRODUCIBILITY

The replication lives in a repository whose instruments were validated under sealed tolerance freezes before any result-bearing run, with every numerical bar fixed in advance. The transcription is guarded by solving the printed linear system numerically against the printed closed form across a parameter sweep, and the provenance note maps each source equation to its implementation line by line. Evidence records are append-only JSON with SHA-256 hashes of inputs and outputs, and every figure in this paper is generated by a committed script whose inputs are the committed evidence records or the committed replication code, with the invariant of Fig. 2 asserted inside the data script itself.

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- [1] E. C. G. Stueckelberg, La signification du temps propre en mécanique ondulatoire, *Helv. Phys. Acta* **14**, 588 (1941).
 - [2] E. C. G. Stueckelberg, La mécanique du point matériel en théorie de relativité et en théorie des quanta, *Helv. Phys. Acta* **15**, 23 (1942).
 - [3] L. P. Horwitz and C. Piron, Relativistic dynamics, *Helv. Phys. Acta* **46**, 316 (1973).
 - [4] M. Land, Pair production in classical Stueckelberg–Horwitz–Piron electrodynamics, *J. Phys.: Conf. Ser.* **845**, 012025 (2017), arXiv:1604.01625.
 - [5] M. Land and L. P. Horwitz, *Relativistic Classical Mechanics and Electrodynamics* (Morgan and Claypool, San Rafael, 2020).
 - [6] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, 2nd ed. (Springer, Berlin, 2006).
 - [7] A. Iserles, H. Z. Munthe-Kaas, S. P. Nørsett, and A. Zanna, Lie-group methods, *Acta Numerica* **9**, 215 (2000).
 - [8] R. P. Feynman, The theory of positrons, *Phys. Rev.* **76**, 749 (1949).